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Network Modeling and Analysis in Python

Today's goal • 3/3 items

Module 1
Connectivity in Networks

Module 2
Community Structure in Networks

Module 3
Network Generative Models

Module 4
Models of Diffusion in Networks

Basic Models of Diffusion in Networks (Part 1)

- Introduction to Diffusion in Networks
  Video • 7 min
- The SIS Model
  Video • 8 min
- The SIR Model
  Video • 6 min
- Knowledge check: Basic Models of Diffusion in Networks (Part 1)
  Practice Assignment • Grade: 100%

Basic Models of Diffusion in Networks (Part 2)

- The Independent Cascade Model
  Video • 5 min
- The Linear Threshold Model (Part 1)
  Video • 8 min
- The Linear Threshold Model (Part 2)
  Video • 5 min
- Knowledge Check: Basic Models of Diffusion in Networks (Part 2)
  Practice Assignment • Grade: 100%

The Influence Maximization Problem

- **Honduras Social Networks**
  Reading • 10 min
- Introduction to the Influence Maximization Problem
  Video • 3 min
- Properties of the Influence Maximization Problem
  Video • 14 min
- Greedy Algorithm for the Influence Maximization Problem
  Video • 4 min
- Heuristics and Nomination Strategy
  Video • 10 min
- Applications of Community Detection
  Video • 7 min
- Knowledge Check: The Influence Maximization Problem
  Practice Assignment • 35 min

Module 4 Wrap-Up

- Demo: Diffusion NetworkX Tutorial

Honduras Social Networks

0:24 / 0:24 2x

This video (Honduras Social Networks) highlights a project by a research group at Yale that uses social network analysis to promote the adoption of beneficial health behaviors in villages in Honduras. As you watch the Honduras Social Networks video, I encourage you to draw connections between the approach and techniques discussed in the video and the concepts we've covered in this course.

Honduras Social Networks (long version)
Yale Institute for Network Science

Watch on

Go to next item ✓ Completed

Video Introduction to the Influence Maximization Problem starts in 5s Start Pause

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